

Program Schedule and Abstract Book

TEST 2026

May 26 - 27, 2026

This is a synthetic example program. All names, titles, and abstracts are fictitious.

Program Schedule

8:00 AM-9:30 AM, Grand Ballroom, PL1. Plenary Session I, Plenary, Presider: Yara Yarrow, Max-Planck-Inst Physik des Lichts

8:00 AM-8:30 AM PL1.1. Phase-stable thin-film platform for beam steering Ferdinand Oakhurst

8:30 AM-9:00 AM PL1.2. Integrated, quantum-limited microresonator enabling optical communication Hester Ivers; Hollis Oosterhout

9:00 AM-9:30 AM PL1.3. Programmable, wavelength-agile microresonator enabling spectroscopy Anselm Holloway

9:45 AM-11:45 AM, W101, TM1A. Integrated Photonic Devices, A&T Oral, Applications & Technology 3, Presider: Cassidy FAIRBANKS, University of Birmingham

9:45 AM-10:15 AM TM1A.1. [Invited Talk] Programmable interferometer for frequency metrology Wren Carrow; Zephyr Fairbanks; Leontine Rothschild; Gunnar Keswick; Imogen Rothschild; Jasper Yamaguchi; Ulrike Jovanovic

10:15 AM-10:45 AM TM1A.2. Toward programmable coherent detection using a optomechanical oscillator Halcyon Drummond; Aurelio Rothschild

10:45 AM-11:15 AM TM1A.3. Coherent, dispersion-engineered thin-film platform enabling photonic computing Ulrike Ivers; Imogen Carrow; Joaquin Galbraith

11:15 AM-11:45 AM TM1A.4. Demonstration of a single-photon spectrometer in spectroscopy Isolde Pemberton; Kasimir Norrington; Quill Rasmussen

9:45 AM-11:45 AM, W102, TM1B. Quantum Light Sources, S&I Oral, Science and Innovations 5, Presider: Rosalind Sandoval, Arizona State University

9:45 AM-10:15 AM TM1B.1. [Invited Talk] Monolithic, programmable Bragg grating enabling neuromorphic computing Dmitri Rothschild; Ulrike Greenleaf; Octavia Sandoval

10:15 AM-10:45 AM TM1B.2. Toward dissipative terahertz imaging using a waveguide lattice Esperanza Stillwell

10:45 AM-11:15 AM TM1B.3. Chip-scale fiber amplifier for coherent detection Nikolai Drummond

11:15 AM-11:45 AM TM1B.4. Abstract Withdrawn

9:45 AM-11:45 AM, W103, TM1C. Ultrafast Phenomena, FS Oral, Fundamental Science 2, Presider: Briony Ashby, University of Sydney

9:45 AM-10:15 AM TM1C.1. [Invited Talk] Hybrid spectrometer for quantum sensing Renata Quintero; Greta Keswick; Per Saintclair; Cassidy Galbraith; Zelda Lammergeier; Octavia Xanthos

10:15 AM-10:45 AM TM1C.2. Toward resonant mid-infrared generation using a laser source Tarquin Brandt; Soren Stillwell; Avery Pendleton; Jericho Tikhomirov

10:45 AM-11:15 AM TM1C.3. Programmable quantum emitter for LiDAR ranging Nikolai Fairbanks; Halcyon Ashby; Isolde Kingsley; Octavia Ophuls; Lucian Eastwick

11:15 AM-11:45 AM TM1C.4. Self-referenced, low-noise waveguide lattice enabling terahertz imaging Greta Jovanovic; Hollis Jardine; Aurelio Zhukova; Kasimir Quintero

9:45 AM-11:45 AM, W104, TM1D. Nonlinear Frequency Conversion, S&I Oral, Science and Innovations 8, Presider: Kasimir Quintero, University of Maryland

9:45 AM-10:15 AM TM1D.1. [Invited Talk] Toward monolithic beam steering using a interferometer Quill Farrow; Caius Gallagher; Lucian Ueno; Esperanza Wojciechowski; Lucian Gallagher

10:15 AM-10:45 AM TM1D.2. Toward topological nonlinear microscopy using a optical cavity Mirabel Lammergeier; Dorian Jessop; Ulric Bellweather; Hollis Fontaine; Ulrike Norrington; Viggo Amundsen; Ushi Stillwell

10:45 AM-11:15 AM TM1D.3. Toward spatially-multiplexed coherent detection using a photonic crystal Gunnar Tikhomirov; Saffron Eastwick; Ushi Stillwell; Valentin Rothschild; Rosalind Engström; Xenia Quintero; Joaquin Gallagher

11:15 AM-11:45 AM TM1D.4. Toward hybrid precision timing using a phased array Magnus Drummond; Ulrike Whitlock; Gideon Yamaguchi; Wilhelmina Devereux; Avery Abernathy; Beatrix Fontaine

1:30 PM-3:30 PM, W101, TM2A. Optical Sensing Systems, A&T Oral, Applications & Technology 3, Presider: Priya JESSOP, Tamagawa University

1:30 PM-2:00 PM TM2A.1. [Invited Talk] Demonstration of a thermally-tuned Bragg grating in frequency metrology Octavia Fairbanks

2:00 PM-2:30 PM TM2A.2. Demonstration of a integrated photodetector in optical networking Nadia Rasmussen; Calliope Calloway

2:30 PM-3:00 PM TM2A.3. Toward mode-locked photonic computing using a gain medium Dorian Voskuijlen; Liesel Blackwood; Viggo Greenleaf

3:00 PM-3:30 PM TM2A.4. Toward electrically-pumped attosecond science using a ring resonator Anselm Kingsley; Hester Carrow

1:30 PM-3:30 PM, W102, TM2B. Laser Resonator Design, S&I Oral, Science and Innovations 5, Presider: Imogen Underwood, ETH Zurich

1:30 PM-2:00 PM TM2B.1. [Invited Talk] Toward gain-switched biophotonics using a soliton Octavia Vandermeer

2:00 PM-2:30 PM TM2B.2. Demonstration of a chip-scale spectrometer in coherent detection Greta Underwood; Priya Lindqvist; Octavia Rasmussen

2:30 PM-3:00 PM TM2B.3. Gain-switched, electrically-pumped optomechanical oscillator enabling optical communication Bennett Nakashima; Emrys Jessop; Nadia Voskuijlen

3:00 PM-3:30 PM TM2B.4. Toward tunable attosecond science using a feedback loop Gunnar Lindqvist; Kasimir Nakashima

1:30 PM-3:30 PM, W103, TM2C. Coherent Optical Networks, FS Oral, Fundamental Science 2, Presider: Renata Nightingale, NBI

1:30 PM-2:00 PM TM2C.1. [Invited Talk] Toward dispersion-engineered neuromorphic computing using a frequency comb Dmitri Holloway

2:00 PM-2:30 PM TM2C.2. Gain-switched, low-noise beam combiner enabling biophotonics Greta Isaksson; Felix Kowalczyk

2:30 PM-3:00 PM TM2C.3. Phase-stable, nonlinear quantum emitter enabling biophotonics Priya Ivers

3:00 PM-3:30 PM TM2C.4. Toward broadband optical communication using a nanobeam Magnus Hawksworth; Dorian Rothschild; Caius Tikhomirov; Salem Bellweather; Wilhelmina Ellsworth

1:30 PM-3:30 PM, W104, TM2D. Mid-Infrared Technologies, S&I Oral, Science and Innovations 8, Presider: Anselm Voskuijlen, Massachusetts Inst of Tech Lincoln Lab

1:30 PM-2:00 PM TM2D.1. [Invited Talk] Integrated ring resonator for optical communication Ushi Lockhart; Esperanza Tennyson; Halcyon Engström; Wren Castellanos; Wilhelmina Oakhurst

2:00 PM-2:30 PM TM2D.2. Phase-stable entangled-photon pair for optical networking Hollis Yelland; Emrys Lammergeier; Kira Saintclair; Nikolai Ulibarri; Gideon Pietrangeli; Valentin Oosterhout; Calliope Bellweather

2:30 PM-3:00 PM TM2D.3. Toward self-referenced nonlinear microscopy using a photodetector Aurelio Lindqvist; Jericho Lammergeier; Saffron Castellanos; Ushi Tennyson; Kira Blackwood; Soren Lindqvist; Gunnar Engström

3:00 PM-3:30 PM TM2D.4. Toward dispersion-engineered attosecond science using a microresonator Caius Drummond; Per Mortlake

4:00 PM-5:30 PM, W105, SYMP1. Symposium on Emerging Photonic Materials, Symposium, Special Symposia, Presider: Mirabel Montgomery, IMEC

4:00 PM-4:30 PM SYMP1.1. [Invited Talk] Reconfigurable, programmable soliton enabling optical networking Thora Dunmore

4:30 PM-5:00 PM SYMP1.2. [Invited Talk] Toward dissipative nonlinear microscopy using a feedback loop Nikolai Halvorsen; Anselm Ellsworth; Esperanza Zellweger; Calliope Farrow; Calliope Quayle; Yusuf Rothschild; Tarquin Quayle

5:00 PM-5:30 PM SYMP1.3. [Invited Talk] Topological modulator for precision timing Octavia Xanthos; Salem Xanthos; Delphine Pietrangeli; Nadia Yelland; Fiora Lindqvist; Renata Carrow; Quentin Nakashima

5:30 PM-7:00 PM, Exhibit Hall, JTUP. Poster Session I, Poster

5:30 PM-7:00 PM JTUP.1. Phase-stable waveguide lattice for precision timing Elena Pietrangeli; Felix Blackwood; Fiora Hawksworth; Nikolai Oosterhout

5:30 PM-7:00 PM JTUP.2. Toward reconfigurable terahertz imaging using a nanobeam Imogen Nakashima; Saffron Lockhart; Gunnar Kowalczyk

5:30 PM-7:00 PM JTUP.3. Demonstration of a resonant spectrometer in beam steering Dorian Fontaine

5:30 PM-7:00 PM JTUP.4. Toward gain-switched optical networking using a quantum emitter Isolde Brandt

5:30 PM-7:00 PM JTUP.5. Spatially-multiplexed, electrically-pumped interferometer enabling beam steering Quill Lockhart; Briony Halvorsen; Valentin Qureshi; Jericho Eastwick; Quill Qureshi

5:30 PM-7:00 PM JTUP.6. Broadband, spatially-multiplexed optical cavity enabling quantum sensing Wilhelmina Whitlock; Cassidy Carrow; Tarquin Amundsen; Kira Dunmore

5:30 PM-7:00 PM JTUP.7. Topological saturable absorber for optical communication Greta Kowalczyk

5:30 PM-7:00 PM JTUP.8. Broadband, polarization-resolved thin-film platform enabling beam steering Yannick Abernathy; Marlowe Ophuls; Per Ulibarri; Dmitri Quayle; Emrys Blackwood
5:30 PM-7:00 PM JTUP.9. Demonstration of a reconfigurable spectrometer in remote sensing Yusuf Pemberton
5:30 PM-7:00 PM JTUP.10. Demonstration of a phase-stable gain medium in spectroscopy Magnus Ellsworth; Marlowe Farrow
5:30 PM-7:00 PM JTUP.11. Coherent, thermally-tuned optical cavity enabling mid-infrared generation Yusuf Yelland; Saffron Lammergeier; Priya Vandermeer; Felix Kowalczyk; Viggo Pendleton
5:30 PM-7:00 PM JTUP.12. Polarization-resolved metasurface for precision timing Yara Ophuls; Zelda Dunmore; Yara Oosterhout

8:30 AM-10:30 AM, W101, WM1A. Photonic Computing Architectures, A&T Oral, Applications & Technology 3, Presider: Wendell KOWALCZYK, UVA

8:30 AM-9:00 AM WM1A.1. [Invited Talk] Toward self-referenced neuromorphic computing using a ring resonator Delphine Quintero
9:00 AM-9:30 AM WM1A.2. Toward monolithic terahertz imaging using a photodetector Caius Saintclair; Ines Pendleton; Tarquin Underwood
9:30 AM-10:00 AM WM1A.3. Wavelength-agile, mode-locked quantum emitter enabling terahertz imaging Ottoline Yarrow; Yannick Lindqvist; Esperanza Wojciechowski; Wilhelmina Pemberton; Halcyon Holloway; Wilhelmina Lindqvist
10:00 AM-10:30 AM WM1A.4. Toward spatially-multiplexed quantum key distribution using a feedback loop Perpetua Jovanovic; Isolde Ashby; Aurelio Abernathy

8:30 AM-10:30 AM, W102, WM1B. Microresonator Combs, S&I Oral, Science and Innovations 5, Presider: Zephyr Ulibarri, University of Florida

8:30 AM-9:00 AM WM1B.1. [Invited Talk] Demonstration of a self-referenced soliton in nonlinear microscopy Leontine Calloway; Saffron Farrow; Kasimir Yarrow; Tarquin Nakashima; Yara Ellsworth; Emrys Zhukova
9:00 AM-9:30 AM WM1B.2. Nonlinear gain medium for biophotonics Viggo Quintero; Quill Greenleaf; Noor Quayle; Anselm Mortlake; Isolde Mortlake; Ulrike Xanthos
9:30 AM-10:00 AM WM1B.3. Low-noise, coherent photonic crystal enabling quantum sensing Dmitri Nakashima; Priya Pietrangeli; Delphine Engström; Tamsin Oakhurst
10:00 AM-10:30 AM WM1B.4. Abstract Withdrawn

8:30 AM-10:30 AM, W103, WM1C. Fiber Laser Advances, FS Oral, Fundamental Science 2, Presider: Calliope Nakashima, University of Arizona

8:30 AM-9:00 AM WM1C.1. [Invited Talk] Toward dispersion-engineered optical communication using a waveguide lattice Ferdinand Yelland; Avery Saintclair; Xenia Kingsley; Viggo Pendleton; Bennett Yelland; Calliope Stillwell; Jericho Sandoval
9:00 AM-9:30 AM WM1C.2. Monolithic photonic crystal for optical communication Calliope Villanueva; Hester Halvorsen; Anselm Kingsley; Vesna Lockhart
9:30 AM-10:00 AM WM1C.3. Demonstration of a phase-stable spectrometer in beam steering Briony Thornbury; Marlowe Norrington
10:00 AM-10:30 AM WM1C.4. Scalable phased array for frequency metrology Octavia Fairbanks; Yara Tennyson; Gunnar Oakhurst; Ushi Farrow; Esperanza Keswick; Felix Ravenscroft

8:30 AM-10:30 AM, W104, WM1D. Metasurface Engineering, S&I Oral, Science and Innovations 8, Presider: Nikolai Quayle, National Institute of Standards and Technology

8:30 AM-9:00 AM WM1D.1. [Invited Talk] Toward resonant biophotonics using a photonic crystal Ines Pemberton; Kira Brandt; Ushi Ravenscroft; Viggo Qureshi; Romilly Bellweather; Fiora Saintclair
9:00 AM-9:30 AM WM1D.2. Resonant, low-noise beam combiner enabling quantum key distribution Hester Underwood; Ottoline Hawksworth; Aurelio Lockhart; Ferdinand Greenleaf; Delphine Ulibarri
9:30 AM-10:00 AM WM1D.3. Quantum-limited, low-noise laser source enabling quantum sensing Dorian Ueno; Octavia Zellweger
10:00 AM-10:30 AM WM1D.4. Toward electrically-pumped optical communication using an entangled-photon pair Saffron Jovanovic; Thora Marchetti; Halcyon Castellanos; Fiora Farrow; Avery Rasmussen; Vesna Holloway; Briony Fairbanks

10:30 AM-12:00 PM, W101, WM2A. Optical Systems, A&T Oral, Applications & Technology 3, Presider: Zephyr ASHBY, NIST/UMD

10:30 AM-11:00 AM WM2A.1. Reconfigurable gain medium for quantum key distribution Esperanza Pendleton; Octavia Drummond

11:00 AM-11:30 AM WM2A.2. Hybrid entangled-photon pair for LiDAR ranging Soren Ueno; Wendell Hawksworth; Leontine Whitlock; Tarquin Dunmore

11:30 AM-12:00 PM WM2A.3. Demonstration of a monolithic metasurface in mid-infrared generation Kira Hawksworth; Salem Kingsley; Greta Ravenscroft; Ottoline Kingsley; Gunnar Castellanos; Octavia Underwood; Fiora Gallagher

10:30 AM-12:00 PM, W102, WM2B. Optical Systems, S&I Oral, Science and Innovations 5, Presider: Tamsin Nightingale, KAIST

10:30 AM-11:00 AM WM2B.1. Demonstration of a ultrafast phased array in beam steering Lucian Dunmore

11:00 AM-11:30 AM WM2B.2. Chip-scale, polarization-resolved entangled-photon pair enabling remote sensing Ulric Rothschild; Gideon Holloway; Jasper Greenleaf; Vesna Pietrangeli

11:30 AM-12:00 PM WM2B.3. Demonstration of a monolithic photodetector in spectroscopy Ferdinand Stillwell; Priya Farrow; Viggo Norrington; Saffron Underwood; Avery Marchetti; Wilhelmina Devereux

10:30 AM-12:00 PM, W103, WM2C. Optical Systems, FS Oral, Fundamental Science 2, Presider: Fiora Zellweger, Institute for Microelectronics Stuttgart

10:30 AM-11:00 AM WM2C.1. Tunable beam combiner for LiDAR ranging Per Marchetti; Caius Whitlock; Wilhelmina Wojciechowski

11:00 AM-11:30 AM WM2C.2. Mode-locked, thermally-tuned ring resonator enabling optical networking Zelda Drummond; Yusuf Vandermeer

11:30 AM-12:00 PM WM2C.3. Toward reconfigurable precision timing using a microresonator Briony Lockhart; Zephyr Carrow; Jasper Dunmore; Bennett Galbraith; Romilly Rasmussen; Soren Mortlake; Thora Inchbald

10:30 AM-12:00 PM, W104, WM2D. Optical Systems, S&I Oral, Science and Innovations 8, Presider: Tamsin Zhukova, Tsinghua University

10:30 AM-11:00 AM WM2D.1. Demonstration of a hybrid nanobeam in LiDAR ranging Kestrel Underwood; Yara Qureshi

11:00 AM-11:30 AM WM2D.2. Toward wavelength-agile terahertz imaging using a Bragg grating Lucian Farrow

11:30 AM-12:00 PM WM2D.3. Demonstration of a self-referenced optomechanical oscillator in attosecond science Zelda Zellweger

1:30 PM-3:00 PM, W101, WT1. Tutorials in Quantum Photonics, S&I Oral, Science and Innovations 9, Presider: Wilhelmina Drummond, University of Central Florida

1:30 PM-2:00 PM WT1.1. [Tutorial Talk] Single-photon frequency comb for nonlinear microscopy Calliope Bellweather; Magnus Thornbury; Ottoline Lindqvist; Greta Carrow

2:00 PM-2:30 PM WT1.2. Spatially-multiplexed, thermally-tuned optomechanical oscillator enabling beam steering Halcyon Saintclair; Tamsin Zellweger; Kasimir Quintero; Dorian Voskuilen; Esperanza Ivers; Beatrix Tennyson; Liesel Pemberton

2:30 PM-3:00 PM WT1.3. Demonstration of a ultrafast modulator in spectroscopy Renata Mortlake

3:30 PM-4:30 PM, W101, PD1. Postdeadline Session 1, Postdeadline Oral, Presider: Anselm Ophuls, University of Hong Kong

3:30 PM-4:00 PM PD1.1. Tunable nanobeam for nonlinear microscopy Rosalind Jovanovic; Caius Jardine; Ulric Sandoval; Fiora Yelland; Leontine Quintero

4:00 PM-4:30 PM PD1.2. Demonstration of a few-cycle quantum emitter in LiDAR ranging Ferdinand Ulibarri; Soren Tikhomirov; Hollis Abernathy; Quentin Inchbald; Yannick Ueno

3:30 PM-4:30 PM, W102, PD2. Postdeadline Session 2, Postdeadline Oral, Presider: Yara Hawksworth, RWTH Aachen University

3:30 PM-4:00 PM PD2.1. Toward scalable frequency metrology using an optical cavity Saffron Quayle; Isolde Ivers; Yannick Thornbury; Wendell Wexford; Fiora Zhukova; Hester Devereux

4:00 PM-4:30 PM PD2.2. Polarization-resolved supercontinuum for beam steering Nikolai Devereux; Yara Quayle; Nadia Ueno; Kestrel Lammergeier; Nikolai Jovanovic; Marlowe Jessop

Final ID: PL1.1

Phase-stable thin-film platform for beam steering

Ferdinand Oakhurst¹;

1. Photonics Research Group, INTEC, Ghent University–imec, Ghent, Belgium.

Abstract (35 Word Limit): A polarization-resolved waveguide lattice is demonstrated for remote sensing. The device achieves competitive performance across the operating band.

Final ID: PL1.2

Integrated, quantum-limited microresonator enabling optical communication

*Hester Ivers;*¹, *Hollis Oosterhout;*¹.

1. Physics Department, Technion, Haifa, Israel.

2. School of Physics, Peking University, Beijing, China.

Abstract (35 Word Limit): We present measurements of a coherent fiber amplifier operating in the context of attosecond science. We discuss the trade-offs governing the design and their implications. Results indicate a clear pathway toward practical deployment.

Final ID: PL1.3

Programmable, wavelength-agile microresonator enabling spectroscopy

Anselm Holloway¹;

1. Physics, University of Alabama at Birmingham, Birmingham, AL, United States.

Abstract (35 Word Limit): A broadband saturable absorber is demonstrated for mid-infrared generation. The device achieves competitive performance across the operating band.

Final ID: TM1A.1

Programmable interferometer for frequency metrology

*Wren Carrow;*¹; *Zephyr Fairbanks;*^{1,2}; *Leontine Rothschild;*²; *Gunnar Keswick;*¹; *Imogen Rothschild;*¹; *Jasper Yamaguchi;*^{1,2}; *Ulrike Jovanovic;*²;

1. Swiss Federal Technology Institute of Lausanne, Lausanne, Switzerland.

2. Guangdong Key Laboratory of Integrated Optoelectronics Intellisense, Southern University of Science and Technology, Shenzhen, China.

Abstract (35 Word Limit): A cryogenic ring resonator is demonstrated for spectroscopy. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM1A.2

Toward programmable coherent detection using a optomechanical oscillator

*Halcyon Drummond*¹; *Aurelio Rothschild*²;

1. CREOL, College of Optics and Photonics, University of Central Florida, Orlando, FL, United States.

2. The Chandra Family Department of Electrical and Computer Engineering, University of Texas at Austin, Leander, TX, United States.

Abstract (35 Word Limit): We report a nonlinear optical cavity that enables biophotonics. Our approach reduces system complexity while preserving fidelity. These findings motivate further study of the underlying dynamics.

Final ID: TM1A.3

Coherent, dispersion-engineered thin-film platform enabling photonic computing

Ulrike Ivers;^{1,2}, *Imogen Carrow;*², *Joaquin Galbraith;*¹.

1. School of Physics, Chemistry and Earth Sciences, Adelaide University, Adelaide, SA, Australia.

2. Berkeley Electrical Engineering & Computer Sciences, University of California, Berkeley, Berkeley, CA, United States.

Abstract (35 Word Limit): We report a coherent microresonator that leverages remote sensing. Characterization confirms agreement with the proposed model. The device achieves competitive performance across the operating band.

Final ID: TM1A.4

Demonstration of a single-photon spectrometer in spectroscopy

Isolde Pemberton¹; Kasimir Norrington¹; Quill Rasmussen¹;

1. NP Photonics, Tucson, AZ, United States.

Abstract (35 Word Limit): Here we investigate a resonant photodetector suitable for frequency metrology, reaching a value below 2×10^{-2} cm/decade in our setup. The device achieves competitive performance across the operating band.

Final ID: TM1B.1

Monolithic, programmable Bragg grating enabling neuromorphic computing

*Dmitri Rothschild*²; *Ulrike Greenleaf*^{1,2}; *Octavia Sandoval*¹;

1. Caltech, Pasadena, CA, United States.

2. Research Center for Materials Nanoarchitectonics, National Institute for Materials Science, Tsukuba, Japan.

Abstract (35 Word Limit): Here we investigate a high-finesse soliton suitable for LiDAR ranging. We discuss the trade-offs governing the design and their implications.

Final ID: TM1B.2

Toward dissipative terahertz imaging using a waveguide lattice

Esperanza Stillwell¹;

1. Department of Materials Science and Engineering, The Ohio State University, Columbus, OH, United States.

Abstract (35 Word Limit): We present measurements of a self-referenced thin-film platform operating in the context of optical communication. Our approach reduces system complexity while preserving fidelity. Our approach reduces system complexity while preserving fidelity.

Final ID: TM1B.3

Chip-scale fiber amplifier for coherent detection

Nikolai Drummond¹;

1. Department of Physics and Nuclear Engineering, US Military Academy, West Point, NY, United States.

Abstract (35 Word Limit): Here we investigate a monolithic entangled-photon pair suitable for precision timing. Results indicate a clear pathway toward practical deployment.

Final ID: TM1C.1

Hybrid spectrometer for quantum sensing

Renata Quintero¹; Greta Keswick¹; Per Saintclair¹; Cassidy Galbraith¹; Zelda Lammergerier¹; Octavia Xanthos¹;

1. California Institute of Technology, Pasadena, CA, United States.

Abstract (35 Word Limit): We present measurements of a coherent photodetector operating in the context of precision timing. These findings motivate further study of the underlying dynamics.

Final ID: TM1C.2

Toward resonant mid-infrared generation using a laser source

Tarquin Brandt^{1,2}, *Soren Stillwell*¹, *Avery Pendleton*¹, *Jericho Tikhomirov*²;

1. NIST, Boulder, CO, United States.

2. ICFO – The Institute of Photonic Sciences, Castelldefels, Barcelona, Spain.

Abstract (35 Word Limit): A cryogenic phased array is demonstrated for optical communication. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: TM1C.3

Programmable quantum emitter for LiDAR ranging

Nikolai Fairbanks;¹, Halcyon Ashby;¹, Isolde Kingsley;¹, Octavia Ophuls;¹, Lucian Eastwick;¹,
1. Christian-Albrechts-Universität zu Kiel (Kiel University), Kiel, Germany.

Abstract (35 Word Limit): This work introduces a single-photon modulator as a platform for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: TM1C.4

Self-referenced, low-noise waveguide lattice enabling terahertz imaging

Greta Jovanovic;¹, Hollis Jardine;¹, Aurelio Zhukova;¹, Kasimir Quintero;¹.

1. Perioperative and Critical Care Theme, NIHR Biomedical Research Centre, Southampton, United Kingdom.

Abstract (35 Word Limit): We report a broadband frequency comb that advances attosecond science. Results indicate a clear pathway toward practical deployment. Characterization confirms agreement with the proposed model.

Final ID: TM1D.1

Toward monolithic beam steering using a interferometer

Quill Farrow³; Caius Gallagher³; Lucian Ueno^{1,3}; Esperanza Wojciechowski³; Lucian Gallagher³;

1. Physics, University of North Texas, Denton, TX, United States.

2. University of Wisconsin, Milwaukee,, Milwaukee, WI, United States.

3. Indiana University Bloomington, Elizabethtown, PA, United States.

Abstract (35 Word Limit): This work introduces a low-noise photonic crystal as a platform for precision timing. We discuss the trade-offs governing the design and their implications.

Final ID: TM1D.2

Toward topological nonlinear microscopy using a optical cavity

*Mirabel Lammergeier;¹; Dorian Jessop;¹; Ulric Bellweather;¹; Hollis Fontaine;¹; Ulrike Norrington;¹;
Viggo Amundsen;¹; Ushi Stillwell;¹;*

1. Technion - Israel Institute of Technology, Haifa, Israel.

Abstract (35 Word Limit): We report a integrated entangled-photon pair that accelerates LiDAR ranging. Our approach reduces system complexity while preserving fidelity.

Final ID: TM1D.3

Toward spatially-multiplexed coherent detection using a photonic crystal

Gunnar Tikhomirov¹; Saffron Eastwick¹; Ushi Stillwell¹; Valentin Rothschild¹; Rosalind Engström¹; Xenia Quintero¹; Joaquin Gallagher¹;

1. Time and Frequency Division, National Institute of Standards and Technology, Boulder, CO, United States.

Abstract (35 Word Limit): A thermally-tuned optomechanical oscillator is demonstrated for nonlinear microscopy. The measured figures of merit exceed prior comparable demonstrations. The device achieves competitive performance across the operating band.

Final ID: TM1D.4

Toward hybrid precision timing using a phased array

*Magnus Drummond*¹, *Ulrike Whitlock*², *Gideon Yamaguchi*^{1,2}, *Wilhelmina Devereux*^{1,2}, *Avery Abernathy*^{1,2},
Beatrix Fontaine^{1,2}.

1. ELI-ALPS, Szeged, Hungary.

2. Kochi University of Technology, Kami, Kochi, Japan.

Abstract (35 Word Limit): A broadband ring resonator is demonstrated for LiDAR ranging, reaching a value below 2×10^{-2} s/decade in our setup. Our approach reduces system complexity while preserving fidelity. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM2A.1

Demonstration of a thermally-tuned Bragg grating in frequency metrology

Octavia Fairbanks¹;

1. Los Alamos National Lab, Los Alamos, NM, United States.

Abstract (35 Word Limit): Here we investigate a mode-locked phased array suitable for frequency metrology, reaching a value below 10^{-9} W/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: TM2A.2

Demonstration of a integrated photodetector in optical networking

Nadia Rasmussen¹, Calliope Calloway¹,

1. Laboratoire Interdisciplinaire Carnot de Bourgogne, Dijon, France.

Abstract (35 Word Limit): We present measurements of a gain-switched modulator operating in the context of nonlinear microscopy. We discuss the trade-offs governing the design and their implications.

Final ID: TM2A.3

Toward mode-locked photonic computing using a gain medium

Dorian Voskuijl¹; *Liesel Blackwood*¹; *Viggo Greenleaf*²;

1. Physics, UC San Diego, San Diego, CA, United States.
2. AIM Photonics, Albany, NY, United States.
3. James Franck Institute, University of Chicago, Chicago, IL, United States.

Abstract (35 Word Limit): This work introduces a coherent photonic crystal as a platform for mid-infrared generation. Characterization confirms agreement with the proposed model. These findings motivate further study of the underlying dynamics.

Final ID: TM2A.4

Toward electrically-pumped attosecond science using a ring resonator

Anselm Kingsley;¹; Hester Carrow;¹;

1. RIKEN, Wako, Japan.

Abstract (35 Word Limit): Here we investigate a low-noise gain medium suitable for frequency metrology, reaching a value below 5×10^{-12} Hz/decade in our setup. Our approach reduces system complexity while preserving fidelity. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM2B.1

Toward gain-switched biophotonics using a soliton

Octavia Vandermeer¹.

1. University of Colorado, Boulder, CO, United States.

Abstract (35 Word Limit): We report a polarization-resolved Bragg grating that augments biophotonics, reaching a value below 5×10^{-9} Hz/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: TM2B.2

Demonstration of a chip-scale spectrometer in coherent detection

*Greta Underwood*¹; *Priya Lindqvist*¹; *Octavia Rasmussen*¹;

1. Department of Electrical and Systems Engineering, Washington University in St. Louis, St. Louis, MO, United States.

Abstract (35 Word Limit): Here we investigate a cryogenic thin-film platform suitable for coherent detection.

Characterization confirms agreement with the proposed model. The device achieves competitive performance across the operating band.

Final ID: TM2B.3

Gain-switched, electrically-pumped optomechanical oscillator enabling optical communication

*Bennett Nakashima*²; *Emrys Jessop*^{1,2}; *Nadia Voskuijlen*¹;

1. DEVCOM Armaments Center, U.S. Army, Picatinny Arsenal, NJ, United States.

2. JILA, University of Colorado Boulder, Boulder, CO, United States.

3. NWPU, Xi'an, China.

Abstract (35 Word Limit): We report a single-photon gain medium that underpins spectroscopy. Our approach reduces system complexity while preserving fidelity. The device achieves competitive performance across the operating band.

Final ID: TM2B.4

Toward tunable attosecond science using a feedback loop

Gunnar Lindqvist¹, Kasimir Nakashima¹,

1. Department of Electrical Engineering, Caltech, Pasadena, CA, United States.

Abstract (35 Word Limit): This work introduces a single-photon metasurface as a platform for LiDAR ranging. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2C.1

Toward dispersion-engineered neuromorphic computing using a frequency comb

Dmitri Holloway¹;

1. Electrical Engineering, University of Colorado Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): Here we investigate a programmable laser source suitable for mid-infrared generation. We discuss the trade-offs governing the design and their implications.

Final ID: TM2C.2

Gain-switched, low-noise beam combiner enabling biophotonics

*Greta Isaksson*¹, *Felix Kowalczyk*²,

1. Electrical and Computer Engineering, University of Southern California, Los Angeles, CA, United States.

2. Wyant College of Optical Sciences, University of Arizona, Tucson, AZ, United States.

Abstract (35 Word Limit): We report a quantum-limited microresonator that accelerates terahertz imaging. We discuss the trade-offs governing the design and their implications. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM2C.3

Phase-stable, nonlinear quantum emitter enabling biophotonics

Priya Ivers:¹;

1. University of California, Santa Barbara, Department of Electrical and Computer Engineering, Pasadena, CA, United States.

Abstract (35 Word Limit): This work introduces a quantum-limited entangled-photon pair as a platform for mid-infrared generation. Our approach reduces system complexity while preserving fidelity. Results indicate a clear pathway toward practical deployment.

Final ID: TM2C.4

Toward broadband optical communication using a nanobeam

*Magnus Hawksworth*¹; *Dorian Rothschild*¹; *Caius Tikhomirov*¹; *Salem Bellweather*¹; *Wilhelmina Ellsworth*¹;

1. School of Engineering, Brown University, Providence, RI, United States.

Abstract (35 Word Limit): A resonant entangled-photon pair is demonstrated for optical networking. We discuss the trade-offs governing the design and their implications. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2D.1

Integrated ring resonator for optical communication

*Ushi Lockhart*³; *Esperanza Tennyson*²; *Halcyon Engström*¹; *Wren Castellanos*³; *Wilhelmina Oakhurst*¹;

1. Tsinghua University, Shenzhen, Guangdong, China.

2. Department of Electronic and Electrical Engineering, University College London, London, United Kingdom.

3. Jiangsu Normal University, Xuzhou, China.

Abstract (35 Word Limit): Here we investigate a scalable saturable absorber suitable for quantum key distribution.

The measured figures of merit exceed prior comparable demonstrations. We discuss the trade-offs governing the design and their implications.

Final ID: TM2D.2

Phase-stable entangled-photon pair for optical networking

*Hollis Yelland*¹; *Emrys Lammergeier*²; *Kira Saintclair*^{1,2}; *Nikolai Ulibarri*¹; *Gideon Pietrangeli*²; *Valentin Oosterhout*^{1,2}; *Calliope Bellweather*¹;

1. GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany.

2. Center for Photonics Technology, Virginia Tech, Blacksburg, VA, United States.

Abstract (35 Word Limit): This work introduces a scalable thin-film platform as a platform for frequency metrology. These findings motivate further study of the underlying dynamics. Our approach reduces system complexity while preserving fidelity.

Toward self-referenced nonlinear microscopy using a photodetector

*Aurelio Lindqvist*¹; *Jericho Lammergeier*³; *Saffron Castellanos*^{2,3}; *Ushi Tennyson*²; *Kira Blackwood*^{2,3};
*Soren Lindqvist*¹; *Gunnar Engström*²;

1. Tulane University, New Orleans, LA, United States.

2. Electrical and Computer Engineering, University of Maryland - College Park, College Park, MD, United States.

3. University of Geneva, Geneva, Switzerland.

Abstract (35 Word Limit): We report a dispersion-engineered optomechanical oscillator that supports LiDAR ranging, reaching a value below 2×10^{-12} Hz/decade in our setup. The device achieves competitive performance across the operating band.

Final ID: TM2D.4

Toward dispersion-engineered attosecond science using a microresonator

*Caius Drummond*¹; *Per Mortlake*¹;

1. University of Louisiana at Lafayette, Lafayette, LA, United States.

2. Optics and Photonics, National Central University, Taoyuan, Taiwan.

Abstract (35 Word Limit): A few-cycle Bragg grating is demonstrated for precision timing. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: SYMP1.1

Reconfigurable, programmable soliton enabling optical networking

Thora Dunmore¹;

1. Mechanical Engineering, University of Maryland College Park, College Park, MD, United States.

Abstract (35 Word Limit): Here we investigate a hybrid laser source suitable for spectroscopy, reaching a value below 2×10^{-12} W/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: SYMP1.2

Toward dissipative nonlinear microscopy using a feedback loop

*Nikolai Halvorsen*², *Anselm Ellsworth*¹, *Esperanza Zellweger*¹, *Calliope Farrow*^{1,2}, *Calliope Quayle*²,
*Yusuf Rothschild*², *Tarquin Quayle*^{1,2},

1. Electrical and Computer Engineering, Aarhus University, Aarhus, Denmark.

2. University of Florida, Gainesville, FL, United States.

Abstract (35 Word Limit): We report a chip-scale interferometer that supports quantum sensing, reaching a value below 10^{-12} s/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: SYMP1.3

Topological modulator for precision timing

*Octavia Xanthos*¹; *Salem Xanthos*²; *Delphine Pietrangeli*²; *Nadia Yelland*¹; *Fiora Lindqvist*²; *Renata Carrow*¹; *Quentin Nakashima*¹;

1. Pacific Northwest National Labs, Richland, WA, United States.

2. DPHY, ONERA, Université Paris-Saclay, Palaiseau, France.

Abstract (35 Word Limit): We present measurements of a integrated modulator operating in the context of nonlinear microscopy, reaching a value below 2×10^{-2} cm/decade in our setup. Results indicate a clear pathway toward practical deployment.

Final ID: JTUP.1

Phase-stable waveguide lattice for precision timing

*Elena Pietrangeli*¹; *Felix Blackwood*¹; *Fiora Hawksworth*¹; *Nikolai Oosterhout*¹;

1. DEVCOM US Army Research Laboratory, Adelphi, MD, United States.

Abstract (35 Word Limit): We report a cryogenic laser source that facilitates biophotonics. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.2

Toward reconfigurable terahertz imaging using a nanobeam

*Imogen Nakashima*¹, *Saffron Lockhart*¹, *Gunnar Kowalczyk*¹,

1. Institute for Soldier Nanotechnologies, Massachusetts Institute of Technology, Cambridge, MA, United States.

Abstract (35 Word Limit): We report a wavelength-agile metasurface that accelerates remote sensing. Results indicate a clear pathway toward practical deployment. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.3

Demonstration of a resonant spectrometer in beam steering

Dorian Fontaine¹;

1. Microsystems and Nanotechnology Division, NIST, Gaithersburg, MD, United States.

Abstract (35 Word Limit): This work introduces a mode-locked quantum emitter as a platform for terahertz imaging. The measured figures of merit exceed prior comparable demonstrations. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.4

Toward gain-switched optical networking using a quantum emitter

Isolde Brandt¹;

1. U.S. Naval Research Laboratory, Washington, DC, United States.

Abstract (35 Word Limit): A cryogenic microresonator is demonstrated for nonlinear microscopy. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.5

Spatially-multiplexed, electrically-pumped interferometer enabling beam steering

Quill Lockhart²; Briony Halvorsen²; Valentin Qureshi¹; Jericho Eastwick²; Quill Qureshi¹;

1. Bunkyo University, Kanagawa, Japan.

2. Microbial Pathogenesis and Immunology, Texas A&M University, Bryan, TX, United States.

Abstract (35 Word Limit): A wavelength-agile spectrometer is demonstrated for terahertz imaging. We discuss the trade-offs governing the design and their implications. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.6

Broadband, spatially-multiplexed optical cavity enabling quantum sensing

Wilhelmina Whitlock², Cassidy Carrow¹, Tarquin Amundsen^{1,3}, Kira Dunmore^{1,3},

1. Oak Ridge National Laboratory, Oak Ridge, TN, United States.
2. University of Massachusetts, Boston, Boston, MA, United States.
3. University of Göttingen, Göttingen, Germany.

Abstract (35 Word Limit): Here we investigate a topological frequency comb suitable for remote sensing. Our approach reduces system complexity while preserving fidelity. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.7

Topological saturable absorber for optical communication

Greta Kowalczyk¹.

1. IMB-CNM (CSIC), Instituto de Microelectronica de Barcelona, Barcelona, Bellaterra, Spain.

Abstract (35 Word Limit): A integrated metasurface is demonstrated for terahertz imaging, reaching a value below 2×10^2 Hz/decade in our setup. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.8

Broadband, polarization-resolved thin-film platform enabling beam steering

*Yannick Abernathy*¹; *Marlowe Ophuls*¹; *Per Ulibarri*¹; *Dmitri Quayle*¹; *Emrys Blackwood*¹;

1. Physics, University of Connecticut, Storrs, CT, United States.

2. Augustana College, Rock Island, IL, United States.

Abstract (35 Word Limit): A single-photon entangled-photon pair is demonstrated for frequency metrology, reaching a value below 2×10^{-3} cm/decade in our setup. These findings motivate further study of the underlying dynamics. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.9

Demonstration of a reconfigurable spectrometer in remote sensing

Yusuf Pemberton¹;

1. Joint Quantum Institute, University of Maryland and National Institute of Standards and Technology, College Park, MD, United States.

Abstract (35 Word Limit): This work introduces a cryogenic phased array as a platform for quantum key distribution, reaching a value below 5×10^{-2} Hz/decade in our setup. Characterization confirms agreement with the proposed model. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.10

Demonstration of a phase-stable gain medium in spectroscopy

*Magnus Ellsworth*¹; *Marlowe Farrow*¹.

1. University of Michigan-Dearborn, Taylor, MI, United States.

2. Leonardo, Rome, Italy.

Abstract (35 Word Limit): This work introduces a monolithic frequency comb as a platform for neuromorphic computing, reaching a value below 10^{-2} cm/decade in our setup. Results indicate a clear pathway toward practical deployment. Characterization confirms agreement with the proposed model.

Final ID: JTUP.11

Coherent, thermally-tuned optical cavity enabling mid-infrared generation

Yusuf Yelland¹; *Saffron Lammergeier*¹; *Priya Vandermeer*¹; *Felix Kowalczyk*¹; *Viggo Pendleton*¹;

1. University of Texas, Austin, Austin, TX, United States.

Abstract (35 Word Limit): This work introduces a broadband spectrometer as a platform for terahertz imaging, reaching a value below 10^3 W/decade in our setup. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.12

Polarization-resolved metasurface for precision timing

Yara Ophuls¹; Zelda Dunmore¹; Yara Oosterhout¹;

1. Tokyo metropolitan university, Hino, Tokyo, Japan.

Abstract (35 Word Limit): This work introduces a thermally-tuned quantum emitter as a platform for frequency metrology. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM1A.1

Toward self-referenced neuromorphic computing using a ring resonator

Delphine Quintero¹;

1. Institute of Applied Physics, The faculty of science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Rehovot, Israel.

Abstract (35 Word Limit): This work introduces a mode-locked microresonator as a platform for remote sensing. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1A.2

Toward monolithic terahertz imaging using a photodetector

Caius Saintclair^{1,2}, Ines Pendleton², Tarquin Underwood².

1. Computer Science and Engineering, New York University, New York, NY, United States.

2. Bridgewater State University, Bridgewater, MA, United States.

Abstract (35 Word Limit): This work introduces a phase-stable thin-film platform as a platform for mid-infrared generation. These findings motivate further study of the underlying dynamics. Characterization confirms agreement with the proposed model.

Final ID: WM1A.3

Wavelength-agile, mode-locked quantum emitter enabling terahertz imaging

Ottoline Yarrow;¹; Yannick Lindqvist;¹; Esperanza Wojciechowski;¹; Wilhelmina Pemberton;¹; Halcyon Holloway;¹; Wilhelmina Lindqvist;²;

1. Central European Institute of Technology, Brno University of Technology, Brno, Czechia.

2. Physics, The University of Texas at Austin, Austin, TX, United States.

Abstract (35 Word Limit): This work introduces a nonlinear metasurface as a platform for biophotonics.

Characterization confirms agreement with the proposed model. We discuss the trade-offs governing the design and their implications.

Final ID: WM1A.4

Toward spatially-multiplexed quantum key distribution using a feedback loop

Perpetua Jovanovic;¹, *Isolde Ashby*;^{1,2}, *Aurelio Abernathy*;^{1,2}.

1. University of California Los Angeles, Division of Physical Sciences, College of Letters and Science, Los Angeles, CA, United States.

2. Physics, University of Arizona, Tucson, AZ, United States.

Abstract (35 Word Limit): Here we investigate a reconfigurable photodetector suitable for biophotonics. These findings motivate further study of the underlying dynamics.

Final ID: WM1B.1

Demonstration of a self-referenced soliton in nonlinear microscopy

*Leontine Calloway*²; *Saffron Farrow*²; *Kasimir Yarrow*²; *Tarquin Nakashima*²; *Yara Ellsworth*¹; *Emrys Zhukova*²;

1. Department of Chemistry, North Carolina Agricultural and Technical State University, Greensboro, NC, United States.

2. State Key Lab of CAD&CG, Zhejiang University, Hangzhou, China.

Abstract (35 Word Limit): A dissipative nanobeam is demonstrated for spectroscopy, reaching a value below 10^{-12} cm/decade in our setup. Results indicate a clear pathway toward practical deployment.

Final ID: WM1B.2

Nonlinear gain medium for biophotonics

Viggo Quintero^{1,2}, *Quill Greenleaf*³, *Noor Quayle*¹, *Anselm Mortlake*^{1,2}, *Isolde Mortlake*³, *Ulrike Xanthos*³;

1. Electrical and Computer Engineering, North Carolina State University, Raleigh, NC, United States.

2. LIGENTEC France, Corbeil-Essonnes, France.

3. GLOBALFOUNDRIES, Malta, NY, United States.

Abstract (35 Word Limit): We present measurements of a polarization-resolved soliton operating in the context of attosecond science. Characterization confirms agreement with the proposed model. Results indicate a clear pathway toward practical deployment.

Final ID: WM1B.3

Low-noise, coherent photonic crystal enabling quantum sensing

Dmitri Nakashima³, Priya Pietrangeli³, Delphine Engström², Tamsin Oakhurst¹,

1. Laboratory for Telecommunication Sciences, College Park, MD, United States.

2. Purdue University, West Lafayette, IN, United States.

3. Electrical and Computer Engineering, University of Florida, Gainesville, FL, United States.

Abstract (35 Word Limit): A chip-scale phased array is demonstrated for LiDAR ranging, reaching a value below 10^{-3} W/decade in our setup. The measured figures of merit exceed prior comparable demonstrations. These findings motivate further study of the underlying dynamics.

Final ID: WM1C.1

Toward dispersion-engineered optical communication using a waveguide lattice

Ferdinand Yelland^{1,2}; *Avery Saintclair*²; *Xenia Kingsley*²; *Viggo Pendleton*^{1,2}; *Bennett Yelland*^{1,2}; *Calliope Stillwell*¹; *Jericho Sandoval*¹.

1. Qaleido Photonics, Shenzhen, Guangdong, China.

2. AV Incorporated, Albuquerque, NM, United States.

Abstract (35 Word Limit): This work introduces a self-referenced nanobeam as a platform for frequency metrology. Results indicate a clear pathway toward practical deployment. The device achieves competitive performance across the operating band.

Final ID: WM1C.2

Monolithic photonic crystal for optical communication

Calliope Villanueva;¹, *Hester Halvorsen*;², *Anselm Kingsley*;^{1,2}, *Vesna Lockhart*;^{1,2}.

1. Department of Electrical Engineering, City University of Hong Kong, Hong Kong, Hong Kong.

2. FEMTO-ST, CNRS-UMLP, Besancon, France.

Abstract (35 Word Limit): We present measurements of a spatially-multiplexed phased array operating in the context of remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: WM1C.3

Demonstration of a phase-stable spectrometer in beam steering

*Briony Thornbury*¹; *Marlowe Norrington*¹;

1. Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing, China.

Abstract (35 Word Limit): A polarization-resolved entangled-photon pair is demonstrated for precision timing. Results indicate a clear pathway toward practical deployment.

Final ID: WM1C.4

Scalable phased array for frequency metrology

*Octavia Fairbanks;¹, Yara Tennyson;¹, Gunnar Oakhurst;¹, Ushi Farrow;¹, Esperanza Keswick;¹,
Felix Ravenscroft;¹.*

1. Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, United States.

Abstract (35 Word Limit): We present measurements of a phase-stable metasurface operating in the context of biophotonics. The device achieves competitive performance across the operating band. Results indicate a clear pathway toward practical deployment.

Final ID: WM1D.1

Toward resonant biophotonics using a photonic crystal

*Ines Pemberton*¹; *Kira Brandt*¹; *Ushi Ravenscroft*¹; *Viggo Qureshi*¹; *Romilly Bellweather*²; *Fiora Saintclair*¹;

1. Physics, Duke University, Durham, NC, United States.

2. Department of Optics & Photonics, College of Optics & Photonics, University of Central Florida, Orlando, FL, United States.

Abstract (35 Word Limit): A tunable thin-film platform is demonstrated for mid-infrared generation, reaching a value below 2×10^4 cm/decade in our setup. The device achieves competitive performance across the operating band.

Final ID: WM1D.2

Resonant, low-noise beam combiner enabling quantum key distribution

*Hester Underwood*¹; *Ottoline Hawksworth*¹; *Aurelio Lockhart*¹; *Ferdinand Greenleaf*¹; *Delphine Ulibarri*¹;
1. Nanjing University, Nanjing, JIANGSU , China.

Abstract (35 Word Limit): We report a gain-switched gain medium that accelerates coherent detection. These findings motivate further study of the underlying dynamics.

Final ID: WM1D.3

Quantum-limited, low-noise laser source enabling quantum sensing

Dorian Ueno¹, Octavia Zellweger¹,

1. University of Colorado, Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): This work introduces a ultrafast supercontinuum as a platform for biophotonics, reaching a value below 2×10^{-12} Hz/decade in our setup. Results indicate a clear pathway toward practical deployment. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM1D.4

Toward electrically-pumped optical communication using a entangled-photon pair

*Saffron Jovanovic*¹; *Thora Marchetti*¹; *Halcyon Castellanos*¹; *Fiora Farrow*¹; *Avery Rasmussen*¹;
*Vesna Holloway*¹; *Briony Fairbanks*¹;

1. Menhir Photonics, Dubendorf, Switzerland.

Abstract (35 Word Limit): A resonant fiber amplifier is demonstrated for quantum sensing. We discuss the trade-offs governing the design and their implications. The device achieves competitive performance across the operating band.

Final ID: WM2A.1

Reconfigurable gain medium for quantum key distribution

*Esperanza Pendleton*¹; *Octavia Drummond*¹;

1. Electrical and Computer Engineering, University of Michigan, Ann Arbor, MI, United States.

Abstract (35 Word Limit): A coherent saturable absorber is demonstrated for optical communication, reaching a value below 5×10^{-12} s/decade in our setup. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2A.2

Hybrid entangled-photon pair for LiDAR ranging

Soren Ueno;¹, Wendell Hawksworth;³, Leontine Whitlock;^{2,3}, Tarquin Dunmore;^{1,2}.

1. Electrical, Computer, and Energy Engineering, University of Colorado Boulder, Boulder, CO, United States.

2. Physics and Astronomy, College of Staten Island, Staten Island, NY, United States.

3. UTC Quantum Center, University of Tennessee, Chattanooga, Chattanooga, TN, United States.

Abstract (35 Word Limit): We report a ultrafast laser source that stabilizes remote sensing. The device achieves competitive performance across the operating band.

Final ID: WM2A.3

Demonstration of a monolithic metasurface in mid-infrared generation

Kira Hawksworth^{1,3}; *Salem Kingsley*³; *Greta Ravenscroft*²; *Ottoline Kingsley*^{1,2}; *Gunnar Castellanos*²; *Octavia Underwood*¹; *Fiora Gallagher*¹.

1. Department of Physics and Astronomy, University of Rochester, Rochester, NY, United States.

2. National Institute of Information and Communications Technology, Koganei, Tokyo, Japan.

3. Rensselaer Polytechnic Institute, Troy, NY, United States.

Abstract (35 Word Limit): Here we investigate a polarization-resolved beam combiner suitable for remote sensing. Results indicate a clear pathway toward practical deployment.

Final ID: WM2B.1

Demonstration of a ultrafast phased array in beam steering

Lucian Dunmore;¹,

1. State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China.

Abstract (35 Word Limit): This work introduces a gain-switched nanobeam as a platform for neuromorphic computing. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2B.2

Chip-scale, polarization-resolved entangled-photon pair enabling remote sensing

*Ulric Rothschild*³; *Gideon Holloway*¹; *Jasper Greenleaf*¹; *Vesna Pietrangeli*²;

1. Edward L. Ginzton Laboratory, Stanford, CA, United States.

2. The Computer Sciences Department, University of Wisconsin-Madison, Madison, WI, United States.

3. Institute for Infocomm Research (I2R), Agency for Science, Technology and Research (A*STAR), Singapore, Singapore.

Abstract (35 Word Limit): A reconfigurable nanobeam is demonstrated for neuromorphic computing, reaching a value below 5×10^3 W/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: WM2B.3

Demonstration of a monolithic photodetector in spectroscopy

*Ferdinand Stillwell*¹; *Priya Farrow*¹; *Viggo Norrington*²; *Saffron Underwood*¹; *Avery Marchetti*^{1,2}; *Wilhelmina Devereux*¹;

1. The Chinese University of Hong Kong, Hong Kong, Hong Kong.

2. Institute of Semiconductors, Henan Academy of Sciences, Henan, China.

Abstract (35 Word Limit): Here we investigate a thermally-tuned beam combiner suitable for frequency metrology. Our approach reduces system complexity while preserving fidelity. Results indicate a clear pathway toward practical deployment.

Final ID: WM2C.1

Tunable beam combiner for LiDAR ranging

*Per Marchetti*¹; *Caius Whitlock*¹; *Wilhelmina Wojciechowski*¹;

1. Physics, Photonics, and Optical Engineering, Bridgewater State University, Bridgewater, MA, United States.

2. Postdoctoral Research Associate, NRC, Washington, DC, United States.

3. Fachgebiet Theoretische Elektrotechnik, Brandenburgische Technische Universität Cottbus-Senftenberg, Cottbus, Germany.

Abstract (35 Word Limit): We report a resonant beam combiner that enables photonic computing. These findings motivate further study of the underlying dynamics. Results indicate a clear pathway toward practical deployment.

Final ID: WM2C.2

Mode-locked, thermally-tuned ring resonator enabling optical networking

Zelda Drummond²; Yusuf Vandermeer^{1,2};

1. Electrical and Computer Engineering, Purdue University, West Lafayette, IN, United States.

2. Friedrich Schiller University Jena, Jena, Germany.

Abstract (35 Word Limit): This work introduces a chip-scale microresonator as a platform for beam steering. The device achieves competitive performance across the operating band.

Final ID: WM2C.3

Toward reconfigurable precision timing using a microresonator

Briony Lockhart^{1,2}; *Zephyr Carrow*²; *Jasper Dunmore*¹; *Bennett Galbraith*^{1,2}; *Romilly Rasmussen*²; *Soren Mortlake*²; *Thora Inchbald*¹;

1. Brookhaven National Lab, Upton, NY, United States.

2. School of Information Science and Technology, Harbin Institute of Technology Shenzhen, Shenzhen, Guangdong, China.

Abstract (35 Word Limit): We report a hybrid supercontinuum that demonstrates neuromorphic computing. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2D.1

Demonstration of a hybrid nanobeam in LiDAR ranging

Kestrel Underwood¹, Yara Qureshi¹.

1. Chi 3 Optics, Boulder, CO, United States.

Abstract (35 Word Limit): We report a spatially-multiplexed optical cavity that enables frequency metrology, reaching a value below 10^{-12} Hz/decade in our setup. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2D.2

Toward wavelength-agile terahertz imaging using a Bragg grating

Lucian Farrow¹;

1. Raymond & Beverly Sackler School of Chemistry, Tel Aviv University, Tel Aviv, Israel.

Abstract (35 Word Limit): We present measurements of a programmable gain medium operating in the context of beam steering. Results indicate a clear pathway toward practical deployment. Results indicate a clear pathway toward practical deployment.

Final ID: WM2D.3

Demonstration of a self-referenced optomechanical oscillator in attosecond science

Zelda Zellweger¹;

1. Materials Science and Engineering Graduate Program, University of Massachusetts Amherst, Amherst, MA, United States.

Abstract (35 Word Limit): A dispersion-engineered spectrometer is demonstrated for coherent detection. We discuss the trade-offs governing the design and their implications. These findings motivate further study of the underlying dynamics.

Final ID: WT1.1

Single-photon frequency comb for nonlinear microscopy

*Calliope Bellweather*¹; *Magnus Thornbury*³; *Ottoline Lindqvist*¹; *Greta Carrow*³;

1. Lumina, Cambridge, MA, United States.

2. University of Missouri-Kansas City, Kansas City, MO, United States.

3. Laser Research Center, Vilnius University, Vilnius, Lithuania.

Abstract (35 Word Limit): A integrated photodetector is demonstrated for coherent detection, reaching a value below 5×10^{-9} Hz/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WT1.2

Spatially-multiplexed, thermally-tuned optomechanical oscillator enabling beam steering

*Halcyon Saintclair*¹; *Tamsin Zellweger*¹; *Kasimir Quintero*¹; *Dorian Voskuijlen*¹; *Esperanza Ivers*¹; *Beatrix Tennyson*¹; *Liesel Pemberton*¹;

1. Electrical and Computer Engineering, The Ohio State University, Columbus, OH, United States.

Abstract (35 Word Limit): Here we investigate a high-finesse laser source suitable for photonic computing. We discuss the trade-offs governing the design and their implications. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WT1.3

Demonstration of a ultrafast modulator in spectroscopy

Renata Mortlake¹,

1. Electric Power Board of Chattanooga, Chattanooga, TN, United States.

Abstract (35 Word Limit): We present measurements of a tunable quantum emitter operating in the context of attosecond science, reaching a value below $2^{>3}</sup>$ W/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: PD1.1

Tunable nanobeam for nonlinear microscopy

*Rosalind Jovanovic*²; *Caius Jardine*^{1,2}; *Ulric Sandoval*¹; *Fiora Yelland*¹; *Leontine Quintero*²;

1. Boston University, Boston, MA, United States.

2. Biomedical Engineering, Texas A&M University, College Station, TX, United States.

Abstract (35 Word Limit): A wavelength-agile optomechanical oscillator is demonstrated for coherent detection. Characterization confirms agreement with the proposed model. Our approach reduces system complexity while preserving fidelity.

Final ID: PD1.2

Demonstration of a few-cycle quantum emitter in LiDAR ranging

*Ferdinand Ulibarri*², *Soren Tikhomirov*², *Hollis Abernathy*³, *Quentin Inchbald*¹, *Yannick Ueno*¹;

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Abstract (35 Word Limit): A integrated quantum emitter is demonstrated for frequency metrology. The measured figures of merit exceed prior comparable demonstrations.

Final ID: PD2.1

Toward scalable frequency metrology using a optical cavity

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Abstract (35 Word Limit): We report a reconfigurable waveguide lattice that enables quantum sensing. These findings motivate further study of the underlying dynamics.

Final ID: PD2.2

Polarization-resolved supercontinuum for beam steering

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Abstract (35 Word Limit): A nonlinear nanobeam is demonstrated for spectroscopy. Our approach reduces system complexity while preserving fidelity. Our approach reduces system complexity while preserving fidelity.